

Player Performance Prediction (Group 1)

Team GitHub Repository

<https://github.com/yashs14282001/Group-1-Team-Project.git>

Plan 1. Project specifications.

Topic - Player Performance Prediction Using Machine Learning

2. Research Question / Problem Statement

Cricket analytics has become increasingly important in modern sports due to the growing use of data-driven decision-making in team selection, player evaluation, and match strategy. However, predicting player performance remains a challenging task because player outcomes depend on multiple factors such as recent form, opposition strength, venue conditions, batting position, and match situations.

This project aims to investigate whether machine learning techniques can be used to accurately predict cricket player performance using historical ball-by-ball cricket match data and player statistics.

Research Question

“Can machine learning models accurately predict cricket player performance using historical cricket match data and player statistics?”

3. Objectives

- To collect and preprocess cricket player datasets from the official BCCI website and IPL statistics sources using web scraping techniques.
- To integrate player profile data and performance statistics into a structured dataset for analysis.
- To perform exploratory data analysis (EDA) to identify patterns in player performance across different formats (Test, ODI, T20, IPL).
- To analyse key performance metrics such as runs, strike rate, wickets, economy rate, and batting/bowling styles.
- To compare player performance trends across formats and identify consistent high-performing players.
- To visualise player statistics and performance insights using data visualisation techniques.
- To prepare the dataset for potential predictive modelling and advanced cricket analytics.

4. Goal / Target

The goal of this project is to develop a machine learning-based prediction system capable of estimating cricket player performance, such as runs scored in a match or innings, using historical cricket data.

The project also aims to demonstrate how sports analytics and machine learning can support:

- player analysis,
- team strategy,
- fantasy sports applications,
- and performance evaluation in cricket.

5. Key Performance Indicators (KPI) – Define Success

Technical KPIs

- Successful collection and preprocessing of large cricket datasets.
- Creation of a clean and structured master dataset.
- Successful implementation of multiple machine learning models.
- Improvement in prediction performance through feature engineering.

Model Evaluation KPIs

For regression models:

- R^2 Score
- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)

For classification models (if applicable):

- Accuracy
- Precision
- Recall
- F1-score

Project Success KPIs

- Ability to generate meaningful and consistent player performance predictions.
- Identification of the most influential factors affecting player performance.
- Production of clear visualisations and analytical insights.
- Successful implementation of an end-to-end machine learning workflow.
- Development of a well-documented and reproducible project repository on GitHub.